

Advanced Econometrics

Linear Regression

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Linear regression

- Linear regression is one of the simplest supervised learning algorithms.
- It is used to model a **quantitative dependent variable**.
- Although it may seem rather simple and obvious compared to more modern statistical learning methods, it is still a **useful** and **widely used** modeling method.
- It often serves as a **baseline** for evaluating results obtained with newer approaches.
- Many much more complex methods can be viewed as **generalizations** or **extensions** of linear regression.
- Understanding linear regression is **key** to learning more advanced methods.

Why do we build models?

- Data usually do not “speak for themselves”.
- Models are a way to **describe the most important relationships** between observed variables in a synthetic manner, i.e., using a small number of estimated parameters.

Preliminary remarks

- A model should be tailored to the research objective.
- **In theory**, it should include all factors affecting the phenomenon under study.
- Such a model would, however, be extremely difficult to construct.
- Even if it could be formulated, it would likely be so complex that it would be of little practical use for drawing conclusions.

Initial stage of building a model

- Formulate research objectives.
- Formulate research hypotheses (verbally and econometrically).
- Hypotheses should be **non-trivial** and **verifiable**.
- A poor objective or incorrectly formulated hypothesis will result in larger errors in inference and/or forecasting.
- Model construction should be **supported by theory** or at least by **well-justified intuition** about relevant factors, the direction of relationships, or the functional form of the model.

Linear relationship

- If we study the effect of one variable on another, then mathematically we are looking for a **functional relationship** between these variables.
- The simplest form is a linear function:

$$y_i = \beta_0 + \beta_1 x_i$$

which has two parameters: β_0 (intercept) and β_1 (slope).

- The **intercept** is the point where the function intersects the vertical axis (usually has no direct interpretation).
- The **slope** is most often the target of our analysis. It tells us how y changes when x changes.

Regression line, regression hyperplane

- We call the line describing the relationship the **regression line**.
- We have a line only for a model with an intercept and one explanatory variable.
- With two explanatory variables, the result is a **regression plane**.
- With many variables, we speak of a **regression hyperplane**.

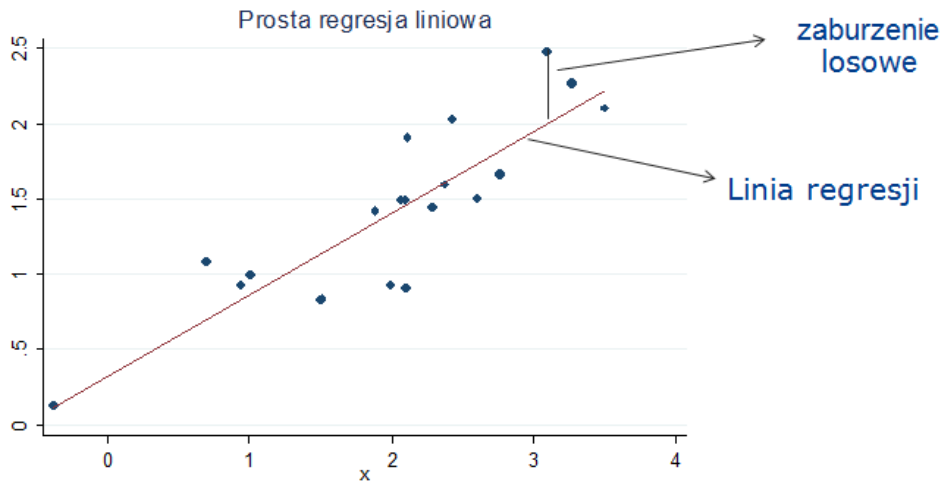
Random disturbance

- In practice, no line fits the data perfectly.
- Therefore, the linear regression function has the following form:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

- The last term (ϵ_i) is the so-called **random component** or **random disturbance**.
- You can think of it as the vertical distance from a data point to the regression line.

Example regression line



Where does the random error come from?

- The **random error** ε_i describes the part of the variability (variance) of y_i that **cannot be explained** using the explanatory variables included in the regression.
- It cannot be explained because:
 - ① the values of other important but **omitted explanatory variables** are unknown,
 - ② some important factors may have a **random nature**.

Simple vs. multiple regression

In simple linear regression:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

- y_i is called: the **explained variable**, dependent variable, endogenous variable, regressand,
- x_i is the **explanatory variable**, independent variable, exogenous variable, regressor.

With many explanatory variables, the regression function is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + \varepsilon_i$$

We then speak of regression of y on $x_1, x_2, \dots, x_k$.

Asymmetry in how variables are treated

- Explanatory variables are treated as **causes** of changes in the dependent variable.
- The direction of causality usually cannot be read directly from the data.
- Such a relationship should follow from theory or intuition:

$$\text{number of beachgoers}_i = \beta_0 + \beta_1 * \text{temperature}_i + \varepsilon_i \implies \text{OK}$$

$$\text{temperature}_i = \beta_0 + \beta_1 * \text{number of beachgoers}_i + \varepsilon_i \implies \text{nonsense}$$

- It is also worth mentioning here the **simultaneity problem**.

Multiple vs. simple regression

- **Main advantage:** multiple regression allows analyzing the relationship between the dependent variable and multiple independent variables simultaneously.
- **Main disadvantage:** increased complexity causes difficulties with:
 - 1 determining which model is “best”,
 - 2 interpreting the estimated parameters.

Estimating a linear regression model

- Before forecasting, we must **estimate** the model parameters from a sample.
- The true parameter values $\beta_0, \beta_1, \beta_2, \dots, \beta_p$ are unknown.
- They are **estimated** based on observed responses of the dependent variable to changes in the explanatory variables in the sample.
- Parameter estimates will of course be **imprecise** to some extent (because they are based on a sample, not the whole population).
- Estimates from two different samples will usually differ.
- In linear regression, estimates are **unbiased** – if we estimated the model on many different samples, the estimates would be **on average** equal to the true parameters.

Least Squares Method

- Parameter **estimate (estimator)** from a sample is usually denoted $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p$ (alternatively $b_0, b_1, b_2, \dots, b_p$).
- The estimator will usually differ from the true parameter value: $\hat{\beta} \neq \beta$; it is an approximation based on the analyzed data sample.
- The value of Y resulting from the estimated model:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \dots + \hat{\beta}_p x_{pi}$$

or in matrix notation:

$$\hat{Y} = X\hat{\beta}$$

is called the **fitted value** or **theoretical value**.

- Theoretical values result from the model and the values of explanatory variables in X .

Residuals

- The difference between the empirical (real, observed) value and the theoretical value is called a **residual**:

$$e_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \dots + \hat{\beta}_p x_{pi}) = y_i - \hat{y}_i$$

or in matrix notation:

$$e = Y - X\hat{\beta} = Y - \hat{Y}$$

- Residuals constitute a **realization (estimate) of the random component**.

Least Squares (LS)

- A good model is one for which residuals are small (predicted values are **close** to the actual values).
- There are many definitions of **closeness**, but the most common is minimizing the sum of squared residuals.
- Residuals can be positive or negative – so we look at their squares (so they don't cancel each other out).
- The smaller the sum of squared residuals, the better the model fits the data.
- We therefore look for estimates $\hat{\beta}$ such that the **sum of squared deviations** of empirical values from theoretical values (residuals) **is as small as possible**.

$$\min_{\hat{\beta}} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \min_{\hat{\beta}} \sum_{i=1}^n e_i^2$$

Form of the LS estimator

- The quantity minimized in LS is called the **residual sum of squares** and is denoted **RSS**.
- It can be shown that in a model with one explanatory variable, the RSS-minimizing parameters are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x}$ and $\bar{y}$ are the sample means of X and Y .

General form of the LS estimator

We can write the general LS estimator as:

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

where X' denotes matrix transposition and the exponent -1 denotes the matrix inverse.

We can estimate it if:

- we have at least as many observations as variables,
- no variable is a linear combination of the others (no multicollinearity).

Statistical properties of the LS estimator

Gauss–Markov Theorem

Under certain assumptions, the LS estimator is the best estimator of the parameter vector among linear and unbiased estimators (***B**est **L**inear **U**nbiased **E**stimator*).

Properties of the LS estimator in CLRM

- ① **Unbiasedness:** an unbiased estimator has expected value equal to the parameter being estimated: $E(b) = \beta$. Intuitively, its estimate is on average equal to the true parameter, so we do not make systematic errors.
- ② **Efficiency:** an efficient estimator has minimal variance (is the most precise).
- ③ **Consistency:** intuitively, as the sample grows, the estimation error goes to 0 and the estimate gets closer to the true value, i.e. $\hat{\theta}_n \xrightarrow{P} \theta$, $n \rightarrow \infty$.

CLRM assumptions

- ➊ There is a **linear relationship** between the dependent variable Y and explanatory variables X (in practice this is often a sufficiently good approximation).
- ➋ Explanatory variables X are **non-random** and do not affect the random errors.
- ➌ The expected value of the random error **equals 0**.
- ➍ Individual random components are **NOT correlated** with each other – **no autocorrelation** of the error term.
- ➎ The variance of the random error is **the same** for all observations – the error term is **homoskedastic**.

CLRM assumptions

- ① $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i$
- ② X deterministic
- ③ $E(\varepsilon_i) = 0$ for $i = 1, \dots, N$
- ④ $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$
- ⑤ $\text{Var}(\varepsilon_i) = \sigma^2$ for $i = 1, \dots, N$ or $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}$

Additional assumption – normality of residuals

- When testing statistical hypotheses in the CLRM, an additional assumption of normality of residuals is adopted:

$$\varepsilon_i \sim NIID(0, \sigma^2)$$

- **Only then** do the statistics used for **hypothesis testing** have the correct theoretical distributions (important in small samples).
- If assumptions are violated, regression results should be treated cautiously and rather as descriptive than as statistical inference.

Interpreting regression parameters

- In a model with many explanatory variables, the estimated parameter tells by how much the **expected value** of the dependent variable Y changes due to a one-unit change in the explanatory variable X associated with that parameter, assuming the other variables remain at the same level (*ceteris paribus*).

$$\beta_k = \frac{\Delta E(Y)}{\Delta X_k}$$

- This is the so-called **partial effect** or **marginal effect**.

Comparing parameter estimates

- A parameter estimate may change if we express an explanatory variable on a different **scale**, e.g., temperature in Celsius vs. Fahrenheit will yield different parameter values.
- Therefore, two parameter estimates in the same model can be directly compared only if the variables they correspond to are measured on the same scale (in the same units).

Standardized parameter estimates

- Comparing parameters for variables measured on different scales can be done only after **standardization**.
- Standardized parameter estimates are obtained by regressing standardized independent variables on the standardized dependent variable.

(Partial) elasticities

- Often, elasticities derived from parameter estimates are more interesting than the estimates themselves.
- A **partial elasticity** is the percentage change in the expected value of Y in response to a 1% change in X_K .

$$e_{Y, X_K} = \frac{\Delta E(y)/E(Y)}{\Delta X_K/X_K}$$

- Their advantage is that they are **dimensionless** (independent of variable units), so they are easier to compare across models.

Elasticity in a linear model

- Elasticity in a linear model **depends on the value of the explanatory variable** and the expected value of Y at which it is computed.

$$e_{Y, X_K} = \frac{\Delta E(y)/E(Y)}{\Delta X_K/X_K} = \frac{\Delta E(Y)}{\Delta X_K} \frac{X_K}{E(Y)} = \beta_K \frac{X_K}{E(Y)}$$

- If we need to compute it, we compute it **at the mean values of the explanatory variables**.
- **In a model where the dependent and explanatory variables are log-transformed, parameters represent elasticities rather than partial effects.**

Testing statistical significance of parameters

- After estimating regression coefficients, we can test their **statistical significance**. Statistical significance means that the **true parameter value differs from zero**.
- Thus we test the **null hypothesis** that the true model coefficient β_i **equals zero**.
- If we **reject** the null hypothesis in favor of the **alternative** that the coefficient is **different from zero**, we say the corresponding variable **is significant** in the model.
- We can write this as: $H_0 : \beta_i = 0$ vs. $H_a : \beta_i \neq 0$.

Testing statistical significance of parameters

- Because we only have a sample, we cannot be certain – we can only test at a chosen **confidence level** (e.g., with probability 95%, 99%).
- Although there is no “hard” rule, one should have at least 20 observations per explanatory variable.
- In general: the more observations, the higher the chance that results will be statistically significant.

Testing statistical significance of parameters

- The **test statistic** has the form:

$$t = \frac{\hat{\beta}_i - 0}{SE(\beta_i)}$$

where $SE(\beta_i)$ is the standard error of the estimator β_i (the estimation error/imprecision due to using a sample rather than the whole population).

- The test statistic t has a Student's t-distribution with $n - 2$ degrees of freedom.
- If the estimation error is **large** and the parameter estimate **small**, we will not be able to reject the null hypothesis.
- Then the nonzero estimated value simply results from estimation imprecision.

Confidence intervals for parameters

- The standard error of a linear regression parameter estimate is proportional to the **residual standard error**: $RSE = \sqrt{\frac{RSS}{n-p-1}}$.
- Based on estimation error, we also compute **confidence intervals** for parameters.
- They show the range within which the true parameter value falls **with high probability** $(1 - \alpha)$.
- α is the **significance level** and is usually small, e.g. 5%, 1%, less often 10%.
- The **larger the standard error**, the wider the confidence interval, and thus the **greater the uncertainty** about the true parameter value.
- Note (from RSE) that the accuracy of parameter estimates increases with sample size (n) .

Testing significance of the entire model

- In addition to testing individual parameter significance, we also test the **overall significance of the model**.
- The null hypothesis is that all **estimated parameters except the intercept** are simultaneously equal to 0: $H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$ vs. the alternative H_a : at least one $\beta_i \neq 0$.
- If the only statistically significant parameter is the intercept, the dependent variable does not depend on the independent variables.
- Then the model cannot explain the variability of the phenomenon.

Testing significance of the entire model – F test

- We use the F test, with the **test statistic**:

$$F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)}$$

where TSS is the total sum of squares, i.e. the sum of squared deviations of Y from its mean: $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$.

- It may happen that the **model as a whole is significant** even though all variables are **individually insignificant**.

Testing joint insignificance of selected variables

- Similarly, we can test the **joint insignificance** of several (e.g., q) **selected variables**:

$$H_0 : \beta_{p-q+1} = \beta_{p-q+2} = \dots = \beta_p = 0$$

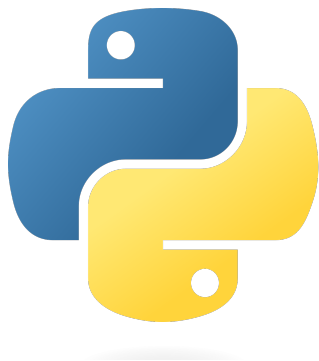
- We then estimate a second model without those q variables.
- If we denote its residual sum of squares as RSS_0 , the test statistic is:

$$F = \frac{(RSS_0 - RSS)/q}{RSS/(n - p - 1)}$$

Statistical tests – reminder

- To decide whether to reject (or not) the null hypothesis based on a test, we need the **null hypothesis** and the **p-value**.
- Roughly, the p-value is the **probability of making an error when rejecting the null hypothesis** – so if it is **low**, we **reject H_0** .
- We compare the p-value with the chosen significance level (α):
 - if $p\text{-value} \leq \alpha$ we **reject the null hypothesis**,
 - if $p\text{-value} \geq \alpha$ we **cannot reject the null hypothesis**.
- The most common significance levels are 5% and 1%.

Hands-on exercises in Python



Unusual observations

Practitioners distinguish three types:

- ➊ **outliers** – observations with a “large” residual, where the dependent variable takes an unusual value given the regressors,
- ➋ observations with a **leverage effect** – where at least one regressor has an unusual value (far from its mean),
- ➌ **influential** observations – removing them leads to substantial changes in parameter estimates (a consequence of the two effects above).

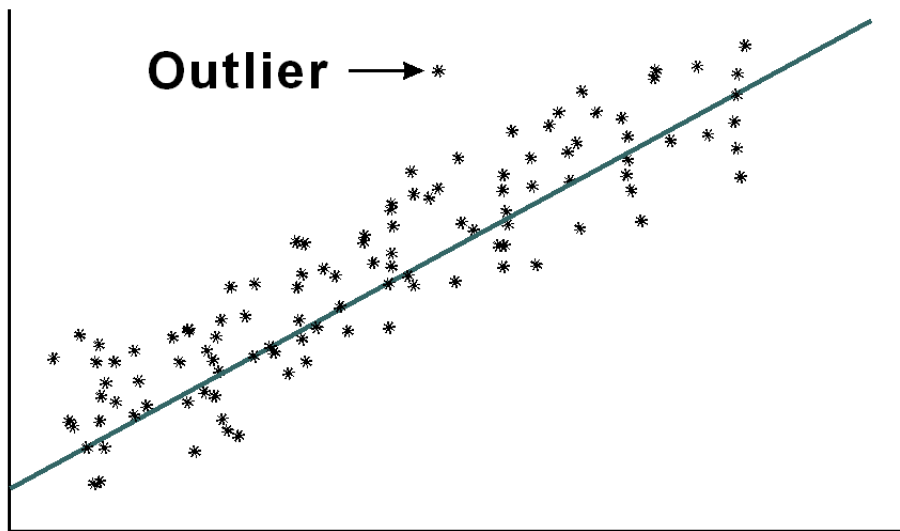
Should we account for them?

- Erroneous observations should not be included in the analysis, but this should be **theoretically justified**.
- It is unacceptable to remove observations solely to improve model fit.
- Unusual observations can be detected using various **descriptive statistics** – but these are not formal statistical tests.
- Detected unusual observations should be checked for correctness, and only then should any decision about removal be made.

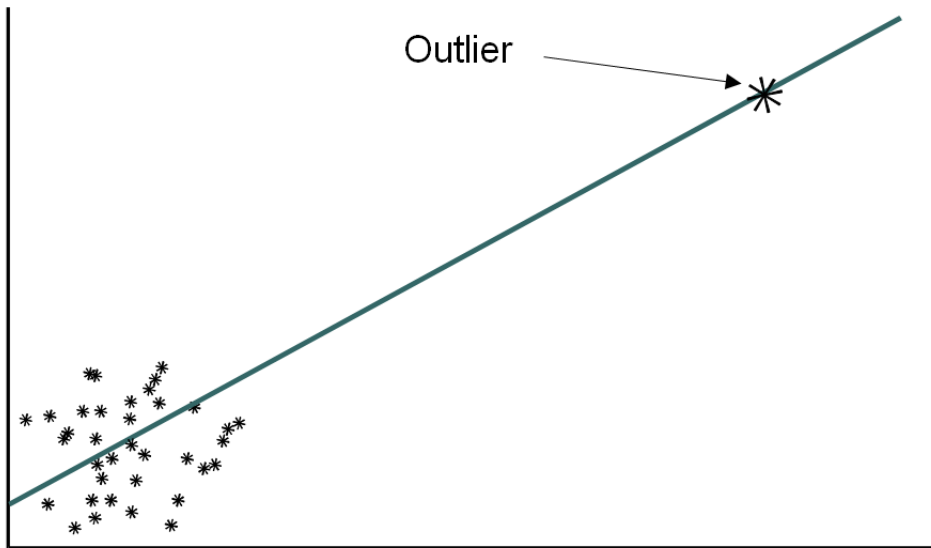
Detecting unusual observations

- Scatter plot
- Frequency tables and contingency tables
- Histogram
- Leverage
- Standardized residuals
- Cook's distance (Cook's D)
- DFFITS statistics
- DFBETA statistics

An unusual observation that does not fit the model



An unusual observation that fits (?) the model



Hat matrix

- The CLRM can be written as: $y = \mathbf{X}\beta + \varepsilon$.
- The estimator of β is $b = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'y$
- Fitted values are: $\hat{y} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'y$.
- Hence we can define the so-called hat matrix:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

- $\mathbf{H}$ is idempotent, symmetric, and has dimensions $(n \times n)$.
- Diagonal elements (h_{ii}) are called **leverage effects** because:

$$\hat{y}_i = h_{ii}y_i + \sum_{i \neq j} h_{ij}y_j$$

Leverage effect

- An observation is considered **unusual** when its leverage satisfies:

$$h_{ii} > \frac{2k}{n}$$

where k is the number of estimated parameters and n is the number of observations used to estimate the model.

- Properties of leverage:
 - ① in any model $0 \leq h_{ii} \leq 1$,
 - ② in a model with an intercept: $\frac{1}{n} \leq h_{ii} \leq 1$.
- The fact that an observation is **unusual** **does not mean it fits the model poorly**. We can check using **standardized residuals**.

Standardized residuals

- Ordinary residuals are:

$$e = y - \hat{y} = y - \mathbf{H}y = (\mathbf{I} - \mathbf{H})y$$

- Since $E(y) = \mathbf{X}\beta$ we can write:

$$e - E(e) = (\mathbf{I} - \mathbf{H})(y - \mathbf{X}\beta) = (\mathbf{I} - \mathbf{H})\varepsilon$$

- It turns out that **ordinary residuals** do not have constant variance:

$$V(e_i) = (1 - h_{ii})\sigma^2$$

so they may be inappropriate for diagnostic purposes.

Standardized residuals

- To address this, we use **standardized residuals**:

$$z_i = \frac{e_i}{\sigma \sqrt{1 - h_{ii}}}$$

- Unfortunately, we do not know σ^2 , the true error variance.

Standardized residuals

- Since the true error variance is unknown, we usually use its estimator:

$$s^2 = \frac{\sum e_i^2}{n - k}$$

where k is the number of parameters to estimate (including the intercept).

- Thus **standardized residuals are ordinary residuals divided by an estimator of the error standard deviation.**
- An observation can be considered unusual when:

$$|r_i^*| > 2$$

- Note that typically there will be relatively many such observations – with many degrees of freedom they make up about 5% of the sample.

Normality of residuals

- In **large samples**, test statistics will have standard distributions even if residuals are not normally distributed.
- In small samples, normality of residuals is required to derive the distributions of test statistics.

Testing normality of residuals

Graphical methods:

- Normal Probability Plot,
- histogram with an overlaid normal density curve.

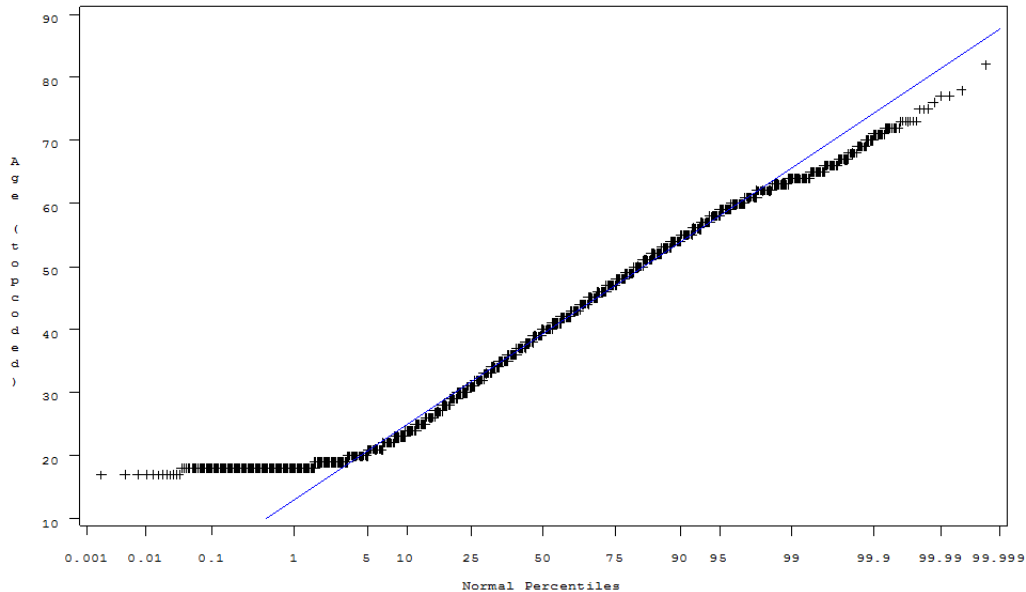
Statistical tests:

- Kolmogorov–Smirnov,
- Anderson–Darling,
- Cramér–von Mises,
- Jarque–Bera,
- Shapiro–Wilk.

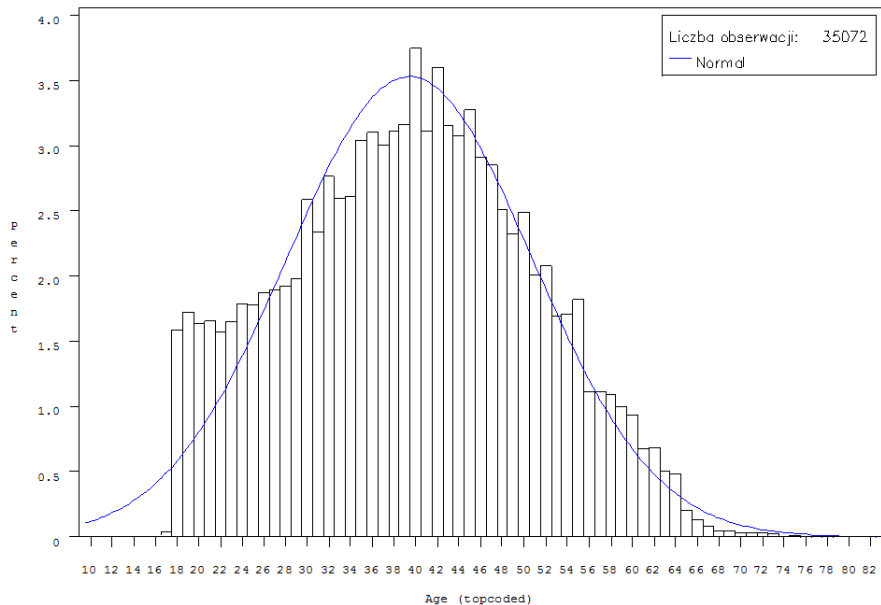
Remedy for non-normal residuals:

- a **log transformation** of regressors is often helpful.

Normal Probability Plot



Histogram with normal density



Multicollinearity (collinearity)

- Multicollinearity means a **strong correlation among explanatory variables**.
- This makes it difficult to determine the individual impact of each independent variable on the phenomenon.
- Two correlated independent variables may be individually insignificant but jointly highly significant.

Perfect multicollinearity

- **Perfect multicollinearity** occurs when one explanatory variable is a **linear combination** of other regressors (columns of $\mathbf{X}$ are linearly dependent).
- Then it is impossible to identify the impact of individual explanatory variables on the dependent variable.

Imperfect multicollinearity

- **Imperfect multicollinearity** occurs when **correlations among regressors are nonzero**.
- This is essentially the rule in real-world data.
- The problem is **strong multicollinearity**.

VIF and tolerance

- Multicollinearity can be detected using the Variance Inflation Factor (*VIF*):

$$VIF_k = \frac{1}{1 - R_k^2}$$

- *VIF* depends on R^2 from regressing the k -th explanatory variable on all remaining regressors.
- High values ($VIF > 10$) suggest strong multicollinearity.
- Sometimes the tolerance coefficient $= \frac{1}{VIF}$ is used; then strong multicollinearity occurs when:

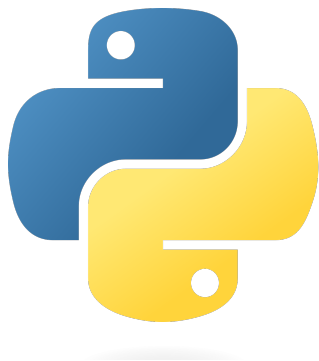
$$\frac{1}{VIF} < \frac{1}{10}$$

- A solution is **removing problematic variables**.

Other ways to assess multicollinearity

- Another approach is computing **condition indices** and corresponding **variance proportions**.
- The condition index is computed as the square root of the ratio of the largest eigenvalue of $X'X$ to each subsequent eigenvalue.
- Rule of thumb:
 - 'CI>10' weak collinearity, 'CI>30' stronger, 'CI>100' severe,
 - for rows with high 'CI', analyze variance proportions,
 - for individual variables: 'VP > 0.5' indicates strong collinearity and identifies a set of collinear variables.

Hands-on exercises in Python



Qualitative explanatory variables

- Explanatory variables often include **qualitative** (discrete) variables.
- These take a finite number of values, usually descriptive in nature.
- A special case is a **binary** (dummy) variable, which takes only two possible values.

Qualitative explanatory variables – binary

- If a variable is binary, it should be **recoded** to 0 and 1 for modeling.
- For clarity, denote a dummy variable in the model as D :

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p D_i + \epsilon_i$$

- Suppose D indicates gender: for men $D = 0$, for women $D = 1$.

Qualitative explanatory variables – binary (cont.)

- Then we can effectively write two separate models (with a small regrouping of β_0):
 - for $D_i = 1$: $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_0 + \beta_p + \epsilon_i$
 - for $D_i = 0$: $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_0 + \epsilon_i$
- β_p shows **how much the intercept differs** for women ($D = 1$) compared to men ($D = 0$).
- In other words, how the **mean value of Y** differs for women compared to men with **the same values of other characteristics**.
- The value $D = 0$ is called the **baseline level** – you can choose arbitrarily which category is coded as 0.

Qualitative explanatory variables – more than 2 levels

- If a **qualitative** variable has **more than two levels**, a single dummy variable cannot represent all levels.
- For example, consider `education_level` coded as: 1 =none, 2 =primary, 3 =secondary, 4 =higher.
- For **each level**, we create a separate dummy variable equal to 1 if the original variable is at that level and 0 otherwise.
- For the example:
 - $pw_1 = 1$ if `education_level` = 1, else 0
 - $pw_2 = 1$ if `education_level` = 2, else 0
 - $pw_3 = 1$ if `education_level` = 3, else 0
 - $pw_4 = 1$ if `education_level` = 4, else 0

Qualitative explanatory variables – more than 2 levels

- We cannot include all these new variables in the model because they are **linearly dependent**.
- Their **sum is always 1**, since each observation must take exactly one category level.
- Then the model always suffers from **multicollinearity**.
- Therefore, we choose one value as the **baseline level** and **do not include** the dummy for that level.

Qualitative explanatory variables – more than 2 levels

- Consider a model where “no education” is the baseline (so pw_1 is omitted):

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p pw_2 + \beta_{p+1} pw_3 + \beta_{p+2} pw_4 + \epsilon_i$$

- Coefficients for category levels represent differences in the mean of y_i **relative to the baseline**.
- For example, β_{p+1} tells how much the expected value of Y differs for someone with **secondary education compared to someone with no education**, holding other characteristics constant.

Interactions

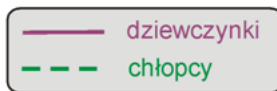
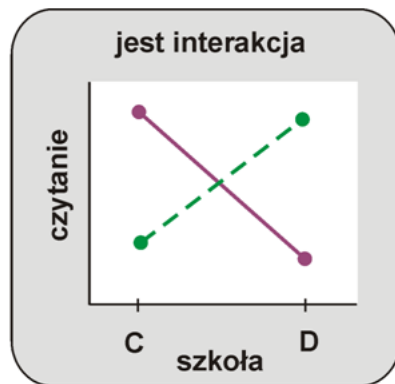
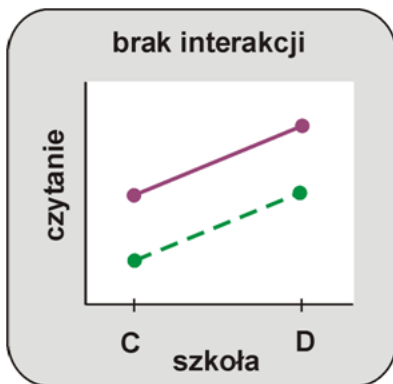
- In a standard linear model we assume effects of variables on the dependent variable are **additive**.
- We can also include **cross-effects** where one variable's impact changes depending on another variable.
- Such an effect occurs when the strength of one independent variable depends on the value of another independent variable.
- This is called an **interaction**.

Interactions

- We add interactions by including explanatory variables created as **products** of variables already in the model.
- Technically, one can add interactions of variables not included in the model, but then the model is not hierarchically well formulated.
- Adding many interactions **can induce multicollinearity**.
- A remedy can be centering variables before forming the interaction.

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \varepsilon_i$$

Interactions



Including nonlinear relationships

- Nonlinearity appears when we try to fit a regression line to data that do not show a “linear shape”.
- A common solution is a **log transformation** of regressors and/or the dependent variable.
- Another solution is **adding higher powers** of explanatory variables (a polynomial model).
- Quadratic polynomial model:

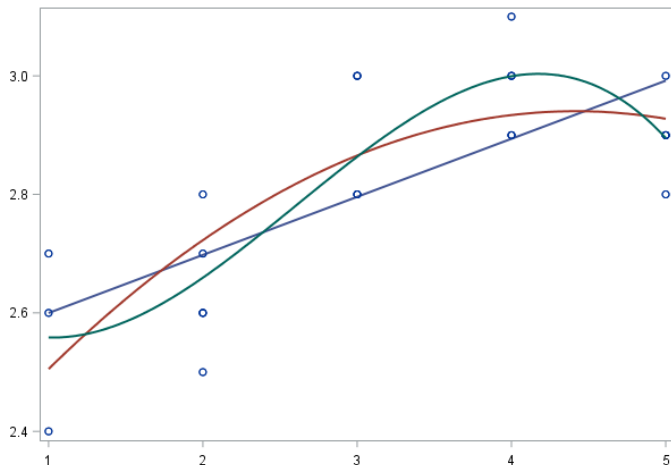
$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i$$

- Cubic polynomial model:

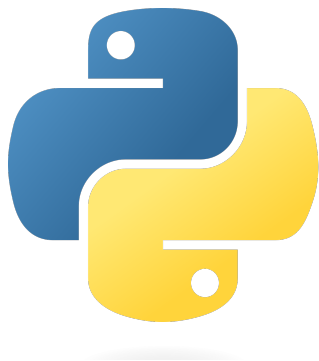
$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \epsilon_i$$

Including nonlinear relationships

- Technically, **nonlinear relationships** are interactions of a variable with **itself**.
- They can therefore cause **multicollinearity problems**:



Hands-on exercises in Python



Variable selection

- Deciding which variables to include is one of the most important issues, especially as data become increasingly high-dimensional.
- As in all of statistics: **garbage in = garbage out**.
- A model with fewer explanatory variables can also be **easier to interpret**.
- If collecting and storing data is costly, it will also be **cheaper to maintain**.
- From a statistical perspective, estimating fewer parameters is **less computationally expensive**.
- In some statistical models, variables that add nothing **can distort results**.
- Using too many variables can also lead to **overfitting**.

Variable selection: supervised vs. unsupervised

- Feature selection can be done **supervised or unsupervised**.
- Unsupervised selection does not use the target variable; variables are chosen based on their individual properties or mutual relationships.
- Supervised selection takes into account **relationships between explanatory variables and the dependent variable**.
- Here variables are chosen to improve accuracy or reduce model complexity.

Omitted vs. irrelevant variables

- Consider two models:
 - (A) $y_i = \beta_0 + \beta_1 x_{1i} + \varepsilon_i$
 - (B) $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$
- **Omitted variables:** the true model is (B) but we estimate (A).
- **Irrelevant variables:** the true model is (A) but we estimate (B).

Omitted vs. irrelevant variables

- From an **econometric perspective**, omitting variables has worse consequences than including irrelevant ones:
 - 1 with irrelevant variables, b is still unbiased but **inefficient** (does not have minimum variance),
 - 2 with omitted variables, b is **usually biased**, meaning we will systematically be wrong.
- On the other hand, from a **prediction-quality** perspective, including extra variables can harm forecasts and reduce predictive power.
- Given the risk of **overfitting** and **poor generalization** to new data, it is often worth limiting the number of variables.

Methods for reducing the number of variables

- In theory, the simplest approach would be to consider **all possible combinations of variables**, compute a chosen quality measure for each model variant, and pick the best one.
- With p variables there are 2^p combinations, e.g., for $p = 40$ we can build over a billion models – **highly inefficient** in practice.
- The solution is variable selection, which can be done in different ways:
 - ① *filter methods* focusing on a **single variable** (or its relationship to the target), independent of the model type,
 - ② *wrapper methods* evaluating many models, adding/removing variables to find an **optimal** combination (according to a criterion).

Filter methods

- Usually **less computationally expensive**.
- We outlined these at the end of the data preparation course; they include correlation analysis, ANOVA models, Wilcoxon tests, t-tests, chi-square tests, and Cramér's V.
- The issue is that their criterion **does not directly refer to the specific model considered**.
- Because filter methods evaluate each variable individually, they may keep redundant variables in the model.

Variable selection methods

- Given current knowledge, this is the **target solution** for high dimensionality that hinders model building.
- In econometric models, **backward, forward, and stepwise** selection are commonly used (and we will cover them today).
- Note that some models are “robust” to irrelevant variables and have **built-in feature selection**, e.g., LASSO regression, decision trees and their derivatives such as random forests – covered in ML courses.

Simple vs. joint hypotheses

- Example: selecting the set of explanatory variables in a model.
- Joint hypothesis:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

with confidence level α .

- Simple hypotheses:

$$H_{01} : \beta_1 = 0$$

$$H_{02} : \beta_2 = 0$$

...

$$H_{0k} : \beta_k = 0$$

where each hypothesis individually uses confidence level α .

Significance level for multiple simple hypotheses

- Based on simple hypotheses, we reject the joint insignificance of k variables if at least one of $H_{01}-H_{0k}$ is rejected.
- These approaches are **not equivalent**.
- If the k test statistics are independent, the **true significance level** is:

$$\alpha^* = 1 - (1 - \alpha)^k$$

- As k increases, α^* approaches 1, meaning the probability of a Type I error becomes very high.
- **Therefore, we should test joint hypotheses.**

Joint hypothesis vs. simple hypotheses

- Testing parameters separately is not equivalent to testing their joint hypothesis.
- **Individually insignificant parameters may be jointly significant.**

General-to-specific approach

- The goal is to determine the correct model form (eliminate irrelevant variables).
- We formulate a sequence of **nested hypotheses**:

$$H_0^0 \subset H_0^1 \subset H_0^2 \subset \dots \subset H_0^K$$

General-to-specific approach

$$H_0^0 : \beta_i \neq 0 \quad \forall \quad i = 1, \dots, K$$

$$H_0^1 : \beta_1 = 0$$

$$H_0^2 : \beta_1 = \beta_2 = 0$$

$$\vdots$$

$$H_0^K : \beta_1 = \dots = \beta_K = 0$$

- H_0^K is nested in H_0^1 and H_0^2 , i.e., it imposes additional restrictions beyond earlier hypotheses.
- Hypotheses $H_0^1, H_0^2, \dots, H_0^K$ are tested sequentially.
- Starting from the most general model (all potential regressors), we move to more specific (simplified) models.

General-to-specific approach

- The final model can differ **depending on the order of hypotheses**.
- Therefore, **the order should have theoretical justification** (e.g., in autoregressive models, reduce higher lags first).

Specific-to-general?

- Can we do the reverse: start with few variables and add more, testing their significance?
- In theory yes, but **with great caution**, and rather only when the reverse method is not available.
- If we omit important variables, statistical inference is incorrect – we cannot properly assess significance of included variables.

Automating variable selection

Criteria for automatic selection:

- rank all possible models e.g. by R^2 ,
- stepwise selection: forward, backward, stepwise.

Forward selection

- 1 Starts with an intercept-only model.
- 2 In each iteration, adds the variable that is most significant.
- 3 Once a variable enters the model, it stays.
- 4 Stops when none of the variables added in the next step is significant at the chosen confidence level.

Backward elimination

- 1 Starts with a model containing all variables.
- 2 In each iteration, removes the least significant variable.
- 3 A removed variable never returns.
- 4 Stops when all variables remaining in the model are significant at the chosen confidence level.

Stepwise selection

- 1 Starts with an intercept-only model.
- 2 In each iteration, adds the most significant variable.
- 3 After adding, removes variables whose significance drops below the chosen threshold.
- 4 Stops when no new variable is significant to add and all variables in the model are significant at the chosen level.

Comparison of selection methods

- Stepwise regression $\implies$ uses fewer computational resources.
- All-subsets $\implies$ offers more choices, but many models may have nearly identical R^2 .

Choosing candidate models (variables)

Candidate models can be selected based on:

- goodness of fit (R^2 , adjusted R^2),
- information criteria (AIC, SBC, HQC),
- variable significance (are all variables significant?),
- diagnostic tests (does the model meet assumptions?),
- **prediction quality** (RMSE, MAPE),
- other criteria.

Decomposition of total sum of squares

- The sum of squared deviations from the mean is a measure of variability:

$$\sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

i.e., total variability (TSS) = explained variability (ESS) + unexplained variability (RSS)

- Possible only in a model with an intercept!

Measures of model fit

- RSS gives an **absolute** measure of misfit.
- But it is measured in units of Y , so we cannot say in general what RSS indicates a good model.
- A better measure is R^2 , the relative counterpart of RSS .
- It takes the form of the **proportion of variance explained**:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

- TSS measures total variability of Y around its mean, and RSS measures the part **not explained** by the model.
- Thus $TSS - RSS$ is the part of variability **explained** by the estimated regression.

Measures of model fit

- R^2 measures the proportion of variability in Y explained by X .
- It always lies between 0 and 1.
- It does not depend on the scale of Y .
- Values closer to 1 generally indicate a better model.
- A value close to 0 may mean:
 - the linear model is incorrect, or
 - estimation error is large (e.g., due to omitted variables).
- There is no universal threshold for a “good” R^2 – it depends on the application.

Measures of model fit

- **Removing a variable** usually **reduces** R^2 .
- **Adding a regressor** (even irrelevant) usually **increases** R^2 .
- That does not mean we should add variables blindly.
- With many variables, use **adjusted** R^2 :

$$adjR^2 = 1 - \frac{n-1}{n-p}(1 - R^2)$$

- Roughly, adjusted R^2 can be interpreted as fit accounting for model complexity.

Notes on regression

- Treat R^2 as descriptive; do not use it as the sole model comparison metric.
- There is no single answer for what R^2 is satisfactory – it depends on the study type.
- Removing a regressor typically worsens R^2 .
- Adding a regressor typically improves R^2 (even if irrelevant) — but don't force variables in!
- Adjusted R^2 accounts for complexity.

Model with intercept or without?

- Models without an intercept should be estimated only when there is a clear theoretical justification.
- Then R^2 does not have the usual interpretation.

Information criteria

- Lower values indicate a better model (note: can be negative).
- Comparable only for models estimated on the same sample.
- Useful also for estimation methods other than LS (e.g., ML) where R^2 is not available.
- In both cases, the first term reflects fit quality and the second penalizes the number of parameters.

Information criteria (cont.)

- Akaike Information Criterion (AIC):

$$\text{AIC} = \ln \left(\frac{\mathbf{e}'\mathbf{e}}{2} \right) + \frac{2K}{n}$$

- Bayesian Information Criterion (BIC/SBC/SIC/SC):

$$\text{BIC} = \ln \left(\frac{\mathbf{e}'\mathbf{e}}{2} \right) + \frac{K \ln n}{n}$$

- Hannan–Quinn Criterion (HQC):

$$\text{HQC} = n \ln \left(\frac{\mathbf{e}'\mathbf{e}}{n} \right) + 2k \ln(\ln n)$$

Forecast error measures

- Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

- Average error magnitude (possible interpretation).
- All observations have equal weight.

Forecast error measures

- Mean Square Error (MSE)

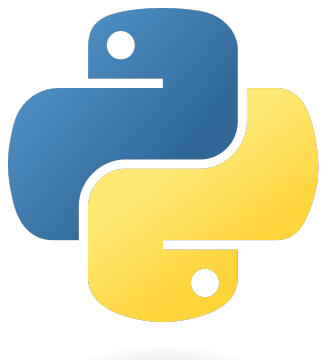
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- Error magnitude without direction (no direct interpretation).
- Observations with large errors receive higher weights.

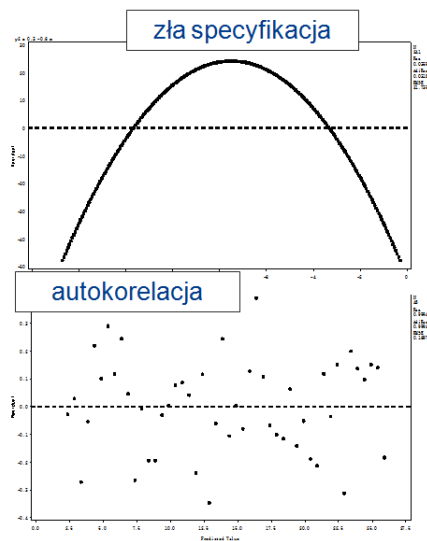
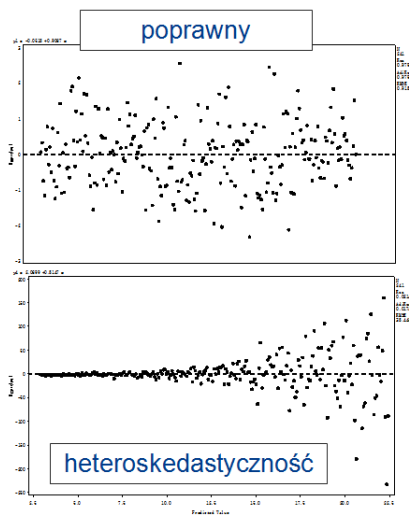
Hands-on exercises in Python



Model diagnostics

- Diagnostic tests verify CLRM assumptions.
- Practically any assumption can be tested.
- Most commonly tested assumptions include:
 - linear functional form,
 - parameter stability,
 - homoskedasticity,
 - no autocorrelation.
- Rejecting CLRM assumptions implies changing the model or estimation method.

Residual scatter plot



RESET test for functional form

- Ramsey's RESET (REgression Specification Error Test) tests model completeness (functional form correctness).
- H_0 : model is correct; no important variables omitted:

$$y_t = x_t\beta + u_t$$

- H_1 : model is incorrect; important variables omitted:

$$y_t = x_t\beta + \sum_{j=2}^p \phi_j \hat{y}_t^j + u_t$$

- If the linear model is correct, the coefficients ϕ_j should not be significant.
- If we reject H_0 , we cannot say anything about the relationship between y and x .

Testing parameter stability - Chow test

- Base model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

- Estimate on disjoint subsamples:

$$\mathbf{y}_1 = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{u}_1$$

$$\mathbf{y}_2 = \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}_2$$

- Null hypothesis:

$$H_0 : \boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$$

- H_0 : no structural break; H_A : structural break exists.

Testing parameter stability - Chow test



Sphericity of the error term

- With autocorrelation or heteroskedasticity, the LS estimator remains unbiased and consistent, but is not efficient:
 - confidence intervals are incorrect,
 - we may draw wrong conclusions about significance,
 - LS residuals are not independent $\implies$ they contain information that can improve predictions.
- A model with autocorrelation or heteroskedasticity has a non-spherical error structure.

White noise process

- A process satisfying homoskedasticity and no autocorrelation is called white noise.
- It is a special case of a stationary process.
- It satisfies:

$$E(y_t) = 0$$

$$\text{Var}(y_t) = \sigma^2$$

$$\text{Cov}(y_t, y_{t-k}) = \begin{cases} \sigma^2 & \text{for } k = 0, \forall t \\ 0 & \text{for } k \neq 0, \forall t \end{cases}$$

Testing white noise

- Box–Pierce statistic:

$$Q = T \sum_{k=1}^m \hat{\rho}_k^2 \sim \chi_m^2$$

- Ljung–Box statistic (better in small samples):

$$Q^* = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \sim \chi_m^2$$

- Notation: T is sample size, m the number of lags, ρ_k the sample autocorrelation.
- Null hypothesis in both tests:

$$H_0 : \rho_1 = \rho_2 = \cdots = \rho_m = 0$$

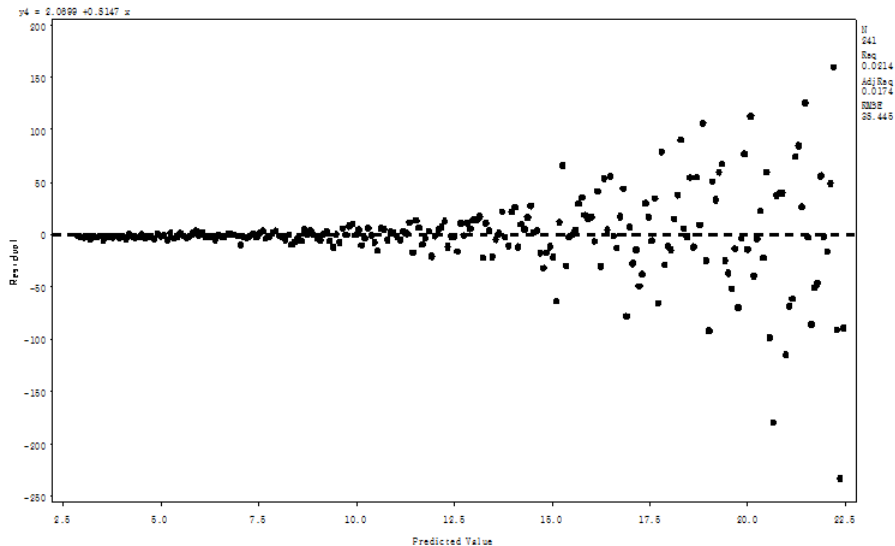
Heteroskedasticity

- Homoskedasticity means constant error variance.
- Heteroskedasticity means the error variance is not constant.
- It often occurs in cross-sectional models.
- Cause: error variance depends on an independent variable.

Detecting heteroskedasticity

- Graphical methods: plot residuals vs. fitted values.
- White test.
- Breusch–Pagan test.

Graphical detection of heteroskedasticity



White and Breusch–Pagan tests

- H_0 : homoskedasticity
- H_A : heteroskedasticity

Procedure:

- Estimate the model and save residuals $\hat{u}_1, \hat{u}_2, \dots, \hat{u}_n$.
- Estimate an auxiliary regression of residuals on explanatory variables (X).
- If fit is sufficiently good, reject H_0 .

Testing for autocorrelation

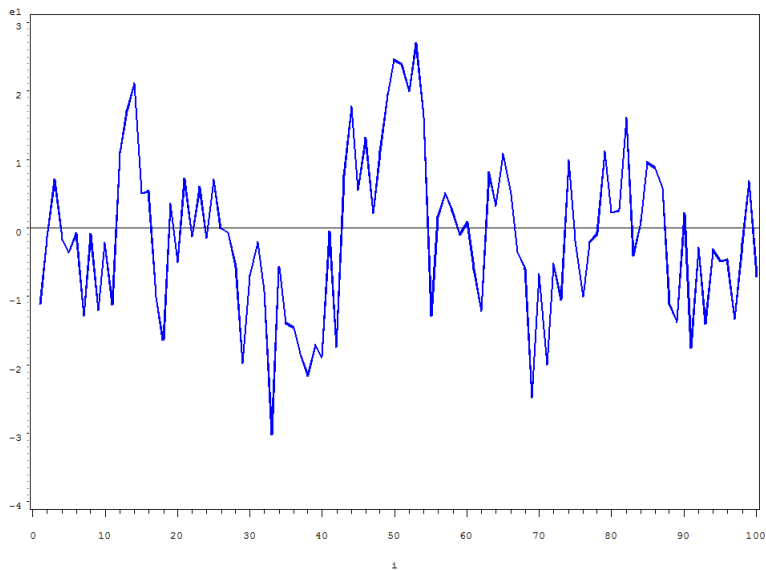
- Dependence of residuals on residuals from other observations.
- Positive autocorrelation often occurs in time-series models.
- Then residuals depend on their lagged values.
- Cause: effects of random events spread over more than one period.
- With misspecified functional form, autocorrelation can also occur in cross-sectional data.

Detecting autocorrelation

- Graphical methods: residual plot.
- Durbin–Watson test.
- Breusch–Godfrey test.

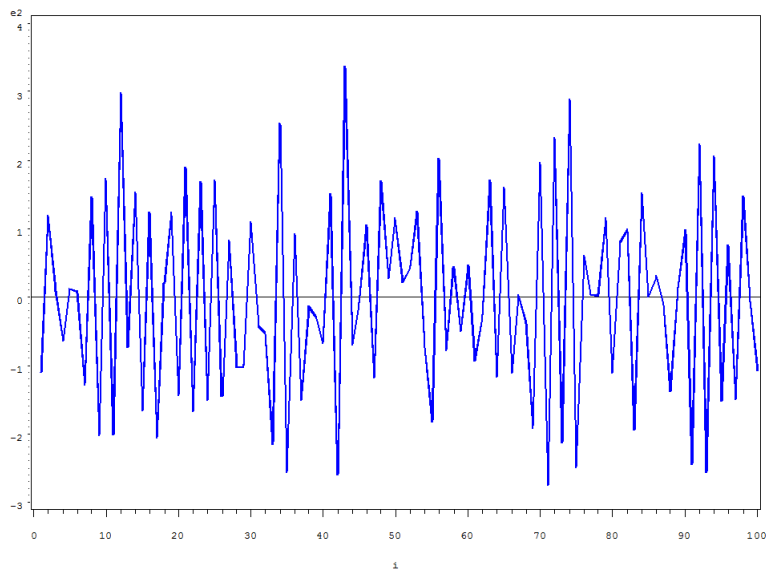
Example of a process with positive autocorrelation

$$y_t = 0.6y_{t-1} + \varepsilon_t$$

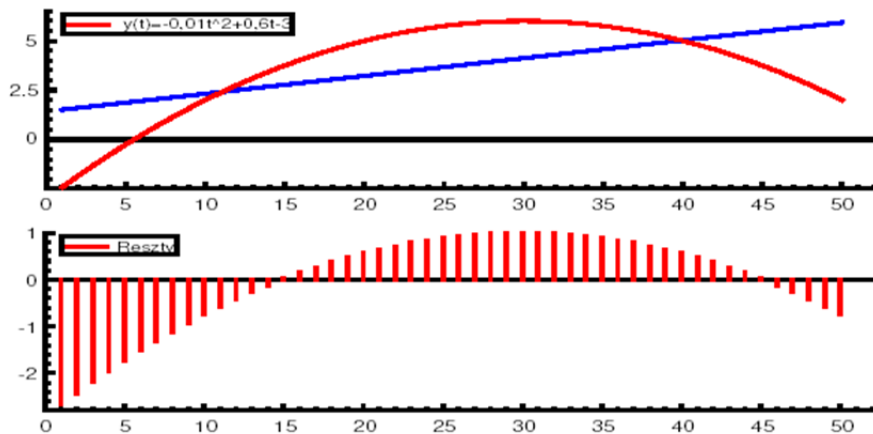


Example of a process with negative autocorrelation

$$y_t = -0.6y_{t-1} + \varepsilon_t$$



Autocorrelation due to misspecification



Durbin–Watson test for residual autocorrelation

- The classic DW test analyzes first-order autocorrelation:

- H_0 : no autocorrelation among model residuals
- H_A : autocorrelation among model residuals

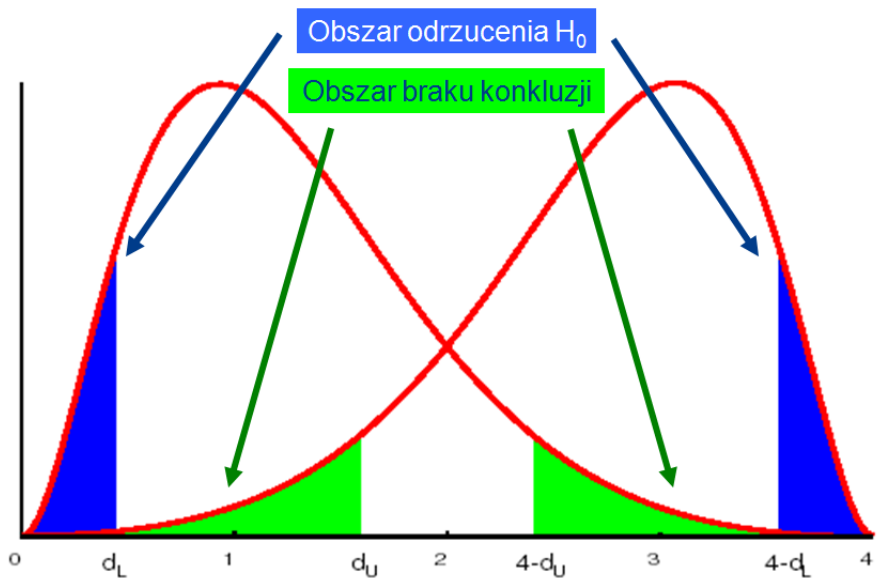
$$e_t = \alpha e_{t-1}$$

- Test statistic $\in [0, 4]$:

$$d = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=1}^T e_t^2}$$

- d close to 2 $\implies$ no autocorrelation.
- d close to 0 $\implies$ positive autocorrelation.
- d close to 4 $\implies$ negative autocorrelation.

Distribution of the Durbin-Watson statistic



Breusch–Godfrey test

Consider the model:

$$Y_t = \beta_1 + \beta_2 X_t + u_t$$

with autocorrelated residuals:

$$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \cdots + \rho_p u_{t-p} + \varepsilon_t$$

Null hypothesis (no autocorrelation):

$$H_0 : \rho_1 = \rho_2 = \cdots = \rho_p = 0$$

Breusch–Godfrey test

Procedure:

- Estimate the model and save residuals.
- Estimate the auxiliary regression:

$$\hat{u}_t = \alpha_1 + \alpha_2 X_t + \rho_1 \hat{u}_{t-1} + \rho_2 \hat{u}_{t-2} + \cdots + \rho_p \hat{u}_{t-p} + \xi_t$$

- Test statistic:

$$(n-p)R^2 \sim \chi_p^2$$

where R^2 is from the auxiliary regression.

What to do with heteroskedasticity and autocorrelation

- White's heteroskedasticity-robust variance–covariance matrix.
- With heteroskedasticity, significance tests and other inference should use robust (White) standard errors.

What to do with heteroskedasticity and autocorrelation

Including autocorrelation structure:

$$y_t = \alpha + \beta x_t + \varepsilon_t$$

where:

$$\varepsilon_t = \phi_1 \varepsilon_{t-1} + \phi_2 \varepsilon_{t-2} + \cdots + \phi_p \varepsilon_{t-p} + \nu_t$$

and:

$$\nu_t \sim N(0, \sigma^2)$$

Thank you for your attention!